

4.2 Overview of Construction

4.2.1 Site Preparation, Grading, Compaction and Drainage

Site preparation includes stripping the site of vegetation and topsoil as indicated on construction documents. Topsoil may be stockpiled and reused for landscaping at the completion of the pavement construction and prior to track surfacing.

Next, depending upon the existing grade, high areas may need to be excavated or low areas filled to establish an appropriate grade. Excavation and/or fill will add to the cost of construction and to the time required for construction. However, it is important to provide a proper grade for the track and surrounding area. The appropriate grade carries the water away from the track and other critical areas while blending the track into its surroundings. Grading is critical to avoiding erosion and the need for excessive ongoing maintenance. In establishing the grade, the acceptable subgrade material must be compacted in accordance with the geotechnical survey. Care must be taken to ensure that the subgrade is compacted uniformly over the entire site.

Proper drainage is critical to ensuring a facility's longevity, including both the subsurface and surface drainage. It is important not to cut costs or take shortcuts when it comes to drainage. Subsurface drainage, if required, usually involves the construction of an interceptor drain within the planned facility. Surface drainage collects and redirects surface water. Water collected in both subsurface and surface drains may be retained on site or moved away from the facility to an approved drainage system. Check local statutory requirements.

For more information on site preparation, excavation and fill, drainage and subgrade construction, see Chapter 2.

4.2.2 Constructing Concrete Curbs and Perimeter Edging

Concrete curbs provide several benefits, help maintain the integrity of the pavement edge and put forth a better finished project. Concrete curbs:

- Strengthen the pavement surface
- Minimize maintenance
- Create a visual border for the track and/or a mow or maintenance strip
- Confine flexible pavement
- Improve subbase compaction
- Control dimensions and elevation grades

For more information on curbs and perimeter edging, see Chapter 2.

4.2.3 Asphalt Guidelines

This guideline is intended to assist owners, contractors, design professionals, and hot mix asphalt suppliers in the design and installation of hot mix asphalt concrete to meet exacting sports construction standards. Because of the wide variation in climate, construction methods, site conditions and materials availability, the services of a qualified and experienced design professional should be obtained before use to ensure suitability for a specific project.

4.2.3a Qualified Suppliers

Hot Mix Asphalt (HMA) shall be manufactured from a state approved/certified HMA manufacturing facility. Work consists of one or more courses of HMA constructed on a properly prepared foundation. The low-volume (aka athletic use) asphalt concrete consists of a mixture of dense (aka well) graded aggregate and specified type and grade of asphalt binder. The manufacturing facility shall be capable of producing HMA in accordance with the following requirements and all applicable local agency specifications on an ongoing and consistent basis.

Ensuring uniform material is produced and selecting the vendor for these asphalt projects will require timely submittal of documents and qualifications to the satisfaction of the owner. Contractor/material supplier shall demonstrate the existence of the following documents:

1. Approved vendor certificate for the locality (state/county/city, et. al.) where work is being done,
2. Quality control manual for material production oversight and testing measures being performed both at the asphalt plant as well as on the job site, and
3. List/organizational chart showing personnel responsible for use of equipment and actions of the crew on the grade while paving and compacting asphalt.

Calibrated equipment and qualified personnel must be always accessible during the construction of the HMA. The contractor installing the HMA shall provide the necessary equipment, materials, and labor to complete the job acceptable to the owner and in accordance with applicable contract documents. Variations in the size and amount of equipment will depend on the size of the area being paved.

It is imperative that all documents list a “Person-in-Charge” who is responsible for the oversight of the previously listed activities. This individual will be the point of contact for the owner and shall work with the owner to ensure timely project completion and specification compliance. This individual shall be knowledgeable in all aspects of asphalt design, production, and installation and shall be an employee of the contractor installing the asphalt, even if the HMA is being produced and supplied by a separate vendor.

This individual shall be knowledgeable in all aspects of asphalt mix design, production, and installation and shall be an employee of the company holding the contract with the owner, even if the HMA is being produced and supplied by a separate vendor. This person shall have authority to take corrective actions needed to make the athletic asphalt fit for use.

For the asphalt installation as well as aggregate base, it is best practice for the owner and or owners’ representatives to review and approve all data and reports prior to the application of athletic surface materials. This includes design submittal, construction observation, and quality control and assurance sampling and testing.

4.2.3b Definitions

1. Surface Course—The low volume surface/wearing course shall be installed uniformly, to all finished lines and grades, smooth, durable, impervious thus protecting lower layers, and stable. Workmanship of the finished surface course shall be of the highest industry standards (NAPA, AI, ASBA, and NHI references) and applicable to sports

surfaces prior to acceptance by the owner. The surface course shall be built with a fine-graded, ½” maximum aggregate particle size (Nominal Maximum Aggregate Size of ¾”) or smaller. Surface course shall be installed to a minimum average 1-½” compacted thickness; ensure that no thickness of less than 1- ¼” compacted thickness is placed.

2. **Leveling Course**—The course and location of the recreational area that requires placement of a variable thickness of HMA to ‘true up’ the area prior to placement of the surface course. This course has a Maximum Aggregate Size (MAS) no greater than that of the surface course.
3. **Base Course**—The lower courses of the pavement structure below the surface and leveling course with a MAS of between ¾” and 1”. Base courses shall not be allowed to remain without the surface course placed over an extended period and as approved by the engineer. The base shall be kept clean and must be completely dry before proceeding. If the minimum thicknesses shown above cannot be met, then install surface mixture as base course.
4. **Tacking/Priming**—The process of applying one coat of emulsified asphalt to all horizontal and vertical surfaces of either an existing pavement for an overlay or between lifts while building an improved or new structure (tacking), or upon the aggregate base (priming).

Designer Note: For other low-volume, fine-graded asphalt mixtures, if permitted by the Engineer, follow the National Center for Asphalt Technology recommended guidelines shown below for minimum lift thicknesses. [Commentary: Absolute minimum is typically achieved by specifying ¼” greater than listed below.]

HMA Mixture NMAAS	NCAT Fine Graded HMA Compacted Thickness, minimum
⅜” (9.5 mm)	(3) x ⅜” = 1- ⅛” (1-½” preferred)
½” (12.5 mm)	(3) x ½” = 1- ½” (2” preferred)
¾” (19.0 mm)	(3) x ¾” = 2- ¼” (3” preferred)

4.2.3c Materials

- Tack Coat and Prime Coat: AASHTO M140 or M208 (Reference the Asphalt Institute MS- 19 for Handling, Storage and Application criteria: typically, 0.05 gal / sy to 0.15 gal / sy (0.02 gal / sy to 0.05 gal / sy residual AC) depending on the existing surface condition; see Surface Preparation section and current AI and NAPA publications).
 - Tack Coat: SS-1, SS-1h, CSS-1 or CSS-1h diluted with an equal amount of water, or agency acceptable product.
 - Prime Coat: Prime Coat materials of MS-2, CMS-2, or HFMS-2s.
- Hot Mix Asphalt (Low Volume):
 - Aggregates, mineral filler, and asphalt binder shall meet or exceed the requirements of local specifications for asphalt pavements placed under this

- contract for qualities and types. The coarse aggregate shall be sound, angular crushed stone, crushed gravel, or crushed air-cooled blast furnace slag (not steel). The fine aggregate shall be well graded, moderately sharp to sharp (angular) sands. No aggregates known to cause rust spots or pop-outs (steel slag, iron pyrite, and / or dust balls) are allowed in the asphalt. No recycled concrete is allowed in any of the asphalt mixtures.
- All HMA mix designs shall be performed in accordance with the Asphalt Institute Manual Series #2 (MS-2), current edition. The HMA mix designs developed shall meet the requirements of one of the following for compactive effort in the laboratory:
 1. Marshall, 50-Blow,
 2. Superpave, 50-Gyratation, or
 3. Hveem, Low Volume Mix.
 4. Alternate **Low Volume** Asphalt Mix Designs may be allowed with the Engineers approval prior to time of bidding.
 - HMA mix designs shall be performed by qualified personnel with proven past experience and successes in the mix design and quality control of asphalt production. Resumes of the signing “individual-in-charge” may be required by the owner and shall be supplied if requested. The design shall meet the following requirements and be less than 24 months old. However, the mix design method used shall be the Contractors option, as stated previously, based on various methods which currently exist around the nation. A completed design shall be signed by a professional engineer and require submittal of documentation as detailed within this specification. This is required by the owner for the producer to demonstrate knowledge of asphalt mix design and production criterion needed to supply athletic asphalt.
 - Bidding documents shall include the Contractors proposed Asphalt Mixture Design sheets. (Reference Mix Design Submittal Checklist in Appendix H.) Designs will be for HMA to be placed for each of the uses anticipated on each project; patching, base, leveling, and/or surface course. Different asphalt suppliers shall require different design submittals. In addition to mix designs, submit Asphalt Placement Work Plan, indicating paving pass widths, paving directions, site access, and timing/coordination of athletic equipment installation (tennis net posts, vault boxes, fencing, etc.). Bid document except the asphalt mix design shall be submitted within 10 days of project award; asphalt mix design cover sheet which summarizes all materials and optimum asphalt content is required with project bidding.
 - All submitted HMA mix designs shall have a completed Mix Design Submittal Checklist (reference Appendix H) and contain at a minimum the following information:
 - All aggregate gradations and quality measurements
 - Plot (0.45 power graph) of final aggregate blend
 - Bulk (dry) specific gravity of all aggregates and final blend (Gsb) including worksheets for natural (virgin) as well as reclaimed asphalt pavement (RAP)
 - Statement of asphalt binder (PG) being used in asphalt mixture
 - Optimum % asphalt binder (Pb)
 - Mix air voids at optimum (Va)
 - Bulk specific gravity of mix at optimum (Gmb)
 - Theoretical maximum specific gravity at optimum (Gmm)

- Voids in the mineral aggregate (VMA) and voids filled with asphalt (VFA)
- Dust to total AC ratio
- All design data and associated design curves

Mix Design Method Requirements

Measures	Superpave	Marshall	Hveem
Stability, lbs.	n/a	1,200 min.	30 min.
Flow, 0.01 in.	n/a	8 to 16	n/a
Swell, in.	n/a	n/a	0.030 max.
Air voids @ optimum AC with 2-hr. aging (cure-time)	3.5%	3.5%	3.5%
VMA based on NMAS			
#4 (4.75 mm)	16.0 min.	16.0 min.	16.0 min.
3/8" (9.5 mm)	15.0 min.	15.0 min.	15.0 min.
1/2" (12.5 mm)	14.0 min.	14.0 min.	14.0 min.
3/4" (19.0 mm)	13.0 min.	13.0 min.	13.0 min.
VFA	73 to 80	73 to 80	73 to 80
Dust to effective AC (design & production)	0.7 to 1.3	0.7 to 1.3	0.7 to 1.3
Tensile strength ratio	80% min.	80% min.	80% min.

In addition, all design methods and measures listed above shall meet the following criteria for design and throughout production.

- Base mixes shall have a minimum of 45% passing the #4 sieve.
- Surface and leveling mixes shall have a minimum of 45% passing the #8 sieve.
- Coarse aggregate fraction shall have a minimum of 85% / 75% crushed faces.
- Fine aggregate angularity (AASHTO T-304, Method A), shall be $\geq 40\%$ with no more than 20% natural sand allowed.
- Mix designs shall include a breakdown factor, increase to minus #200, introduced during the design stage to mimic production values.
- VMA is based on the aggregate bulk (dry) specific gravity, G_{sb} , as determined by AASHTO T-84 and T-85.
- Performance Graded (PG) binder shall meet typical agency specification for new construction low volume roadways (reference LTPPBind, current edition; 98% reliability).
- Reclaimed Asphalt Pavement (RAP):
 - a. May be used up to 20% in the HMA binder/base and leveling courses.
 - b. Typically, 0% for HMA surface course due to potential blemishes forming from unknown or varying aggregate qualities; however, may be used up to 15% in the

- HMA surface course only if approved by the architect/engineer of record.
 - c. Requires a signed and notarized letter stating that no pyrite, steel slag, or aggregates known to rust or deteriorate are within the RAP being utilized is required.
 - d. Requires the Gsb of the RAP to be determined and used in VMA calculations, not the Gse. The RAP Gsb shall be determined after running the RAP Gmm and then from calculating the RAP Gse minus 0.1 for high absorptive aggregates and RAP Gse minus 0.05 for low absorptive aggregates.
 - e. Requires that all RAP shall be crushed and screened over a $\frac{5}{8}$ " screen deck or smaller for binder/base and leveling courses and $\frac{1}{2}$ " screen deck for surface mixtures—no exceptions.
- Reclaimed Asphalt Shingles (RAS) are not allowed in athletic facilities asphalt.
 - Current quality control testing of the mixture, aggregates, and RAP proposed to be used on the project shall be submitted to the owner prior to acceptance of the proposed mix design.

4.2.3d Testing

- Testing required to validate or control the mix supplied is the Paving Contractor's responsibility and will be included in the bid cost for providing these HMA items. Daily maximum theoretical specific gravity (Gmm) values must be made available to the Contractor's density technician for verifying in-place density within four hours of start of production. Asphalt content, gradation, and bulk specific gravity (Gmb) testing shall be performed on the first day of installation for each product used, then done a minimum of once every 400 tons of HMA supplied or every third day for low tonnages that when added together successively do not equal 400 tons. Acceptable average measures are made by use of a correlated nuclear density gauge, a correlated Pavement Quality Indicator or PaveTracker (non-nuclear) or by cutting (4) cores per lift, per day and testing per AASHTO T-166, Method C. Additional testing shall be performed on any given day once 400 tons of asphalt is placed on that day.
- The average sub-lot (daily or 400 tons; whichever is less) in-place density measure for surface course mixtures shall be 94.0% of Gmm with no value less than 92.5% of Gmm. Base and leveling installation of asphalt shall meet local DOT specifications for in-place density measures or average of 92.0% of Gmm, whichever is greater. Surface course longitudinal joints shall be measured directly upon the joint, centered upon by core or density gauge, and shall meet the mat density requirements. Base and leveling course longitudinal joint density measures shall achieve between 95% - 102% of maximum achievable individually, with an average of 98% on any given day.
- Process control testing shall be in accordance with state standards for frequency and methods where the work being performed is done with a minimum of testing meeting the above QC requirements.
- Process Control Voids and minus #200 gradation shall target mix design with no test outside plus/minus 1.0% and VMA shall target the asphalt mix design value or greater, with no test value less than minimum allowed minus 0.3%.
- Print outs of ingredients used shall be supplied for each run of asphalt; data logger or computer screen shot. Print outs shall be supplied daily with the final load of asphalt ticket.

4.2.3e Recommended Means & Methods

Equipment

- Tack/prime coat distributor truck must have an insulated tank, heating system, and a calibrated distributor capable of maintaining a uniform application of emulsified asphalt under pressure throughout the area to be paved. This requires a pump in good working order, full circulating spray bars, and free flowing nozzles. Small, isolated areas may be tacked with a wand.
- Trucks shall have smooth, clean, and tight metal beds that do not have mixture sticking to the truck bed and from which the entire quantity of HMA can be discharged smoothly into the spreading equipment. Trucks shall have a tarp and insulation as needed to protect the asphalt mixture from wind, rain, and cold temperatures. Trucks for hauling asphalt mixture shall be in good, safe working condition.
- Paving equipment must be capable of placing, spreading and finishing courses of HMA to the specified thicknesses. HMA shall be free of marks, segregation and be placed to the required uniform elevation with a smooth texture not showing tearing, shoving, or gouging. Auger extensions are required while pavers are extended beyond the basic screed width. Paving equipment shall be self-propelled and capable of maintaining the line and grade shown on the plans with suitable electronic equipment. The screed shall be straight and true with no bow and utilizing a vibratory screed. Hand work shall be minimized to ensure the best possible finished surface. It is recommended that paving equipment be equipped with sonar pods or no contact skis for sports asphalt construction. Additionally, it should be equipped with automatic slope control to maintain required tolerances. Finally, paving equipment should have fully functional screed heaters and joint preheaters.
- Rollers shall conform to the manufacturer's specifications for all ballasting. At least one vibratory roller shall be required for each project with two rollers required as a minimum. (Three rollers shall be required when tonnage is greater than 300 tons / day.) Rollers shall be of good condition and capable of compacting the HMA to the minimum in-place density required by this specification.

Surface Preparation

1. Repair pavement failures and perform crack repair according to specification requirements prior to HMA installation.
2. Cold-milling and/or grinding may be necessary to ensure that the asphalt edges at concrete abutments such as approaches, sidewalks, curbing, and drainage basins have smooth transitions.
3. After site review, detail whether wedge milling is necessary to assure positive drainage and transition. Install leveling course, if required, on the project per the site details and quantities shown on the plan sheets.
4. Existing surfaces to receive HMA must be clean prior to the installation of any portion of the work. Clean the surface on which the asphalt concrete is to be placed, and keep it free of accumulations of materials that would contaminate the mixture, prevent bonding, or interfere with spreading operations. Methods used may include but not be limited to the use of a sweeper that can wet and vacuum the area free of dirt and debris, clay, and dust, or any other foreign material.

5. Any oil or grease spots shall be scraped and treated to prevent bleeding through the tack coat. Bad oil spills may require removal with a wire brush or other suitable tool. Maintain clean pavements prior to applying emulsified tack coat. When approved sub-grade or pavement courses previously constructed under the Contract become loosened, rutted, or otherwise defective, the Contractor must correct the deficiency according to the contract item or items involved before the spreading of a subsequent pavement course.
6. If subsequent lifts are laid beyond 24 or 48 hours, apply tack coat at the diluted rate of 0.05 gal / sy (0.02 gal / sy residual AC) over newly constructed asphalt leveling or base mixes, 0.10 gal / sy (0.04 gal / sy residual AC) over existing asphalt pavements and 0.15 gal / sy (0.05 gal / sy residual AC) over milled surfaces. The higher rate shall be used on dry and brittle surfaces. All vertical edges abutting proposed asphalt surfaces shall receive a tack coat. Excessive asphalt applications, drooling, or pooling shall be swept with a broom to ensure proper bonding of the HMA. Install the HMA after the asphalt emulsion has 'broken'; i.e. turned from a brown to a black color, indicating water has evaporated. If pick up occurs, wait until emulsion cures.
7. If shown on the plans, apply prime coat at the diluted rate of 0.30 gal / sy over newly placed aggregate base course prior to the installation of the base asphalt.
8. Install tack / prime coat during appropriate weather conditions and protect the tack / prime coat from traffic so as not to wear and track. Allow the tack / prime coat to 'break', i.e. turn from brown to black prior to installation of the HMA.
9. Perform work in appropriate weather conditions that are dry with no rain, snow, or other forms of precipitation falling or imminent (anticipated during installation of the HMA).

Pavement Placement

1. Install HMA which shall generally arrive on the project between 270–300° Fahrenheit (see producer recommendation) asphalt in accordance with above weather conditions and with a temperature of 50° F. and rising for all asphalt lifts.
2. Establish an acceptable rolling pattern with the assistance of a density technician on the first day of construction. Record temperatures, equipment, rolling pattern, and in-place density results throughout the project.
3. Surface course longitudinal joints shall be smooth and true; no deviation from level and true as required of the mat will be allowed. Detail and submit to the owner a paving plan on the site plan sheet prior to placement of asphalt.
4. The entire athletic surface course shall be paved on the same day. The timing and process should be discussed with and approved by the owner before proceeding with the work. If a cold seam will occur, it must be agreed to with the owner in advance, such as near or at a planned saw and seal joint or under the fence line.
5. Rolling shall start as soon as the HMA can be compacted without displacement. Rolling shall continue until the HMA is thoroughly compacted and all roller marks have disappeared. Compact the HMA to a minimum in-place density of 94.0% of the Theoretical Maximum Specific Gravity, Gmm. (Reference density pay factor table.)
6. Smoothness shall meet the requirements of no greater than ¼" in 10 ft. for base and leveling courses and ¼" in 10 ft. for surface course.
7. Thickness of the overall mat shall be within ¼" (surface course, no minus) of the specified plan thickness at all locations. However, the yield for the day and for the entire site shall meet calculated theoretical based on 94% of Gmm supplied from the contractor's mix design and daily test values.

8. Paving joints (install/build control joints per the architectural drawings)

- Minimize construction, longitudinal, and transverse joints left open for an extended period.
- Construct longitudinal joints by paving in a hot fashion with a temperature of not less than 220°F to ensure maximum performance.
- Compact all joints to provide for a neat, uniform and tightly bonded joint that will meet both surface tolerances and density requirements.
- Cut straight and true (vertical) construction or transverse joints if the material has cooled to less than 220°F prior to the placement of the next pass to ensure the best performing joint possible.
- Off-set joints a minimum of 6" between lifts of asphalt.

9. Allow positive drainage off of the athletic facility and towards drainage outlets. Any ponding of water is not acceptable and shall require correction or replacement at the Contractor's expense and as directed by the Engineer. Flood pavement areas as directed by the owner and in the presence of the architect/engineer and surfacing contractor to determine positive drainage acceptability. Review ASBA's "Water on Tracks" position paper for more details.

10. Protect the HMA until such time that coating can be placed upon the properly compacted asphalt, particularly during other construction activities between asphalt installation and athletic surface installation.

11. If excessive segregation is occurring during placement operations, the Contractor will investigate the cause(s) and make appropriate changes to the satisfaction of the owner. (Reference AI MS-22.)

12. Excessive leveling and smoothness correction required to be performed by the surfacing contractor shall be the responsibility of the paving contractor.

Work Timeliness

HMA Full-depth pavements/overlays (a/k/a Resurfacing): The repairs, HMA overlay, and coating shall be accomplished in such a manner as not to unduly delay the progress of the project. Every attempt should be made to complete the surface course placement process in one continuous placement with no cold joints. The timing and process should be discussed with the owner before proceeding with the work.

Site Specific Identification, contractor shall:

- Remove all waste materials from the site and dispose of according to local ordinances.
- Complete all work in compliance with the American Sports Builders Association (ASBA) requirements or as modified here, whichever is more restrictive.
- Notify owner when work is complete.
- Supply owner with Notarized Certificate of Compliance for all products used on the project.
- Supply owner with yield calculations for all products used on the project. (For example, placement of 1,300 sq. yds. of Hot Mix Asphalt, 1-¾" compacted thickness will require 128 tons when the unit weight = 150 pcf.)

References:

- *Asphalt Institute, Lexington, KY [MS-2, Asphalt Mix Design and MS-22, Asphalt Construction.] National Asphalt Pavement Association, Lantham, MD. [Athletic Asphalt.]*
- *National Center for Asphalt Technology [HMA Materials, Mixture Design, and Construction.]*

4.2.3f Asphalt Acceptance

It is the responsibility of the asphalt contractor or the general contractor to supply a base that meets the specifications of the designer and/or the rules of the sport. It is far more difficult and costly to wait until the surface has been installed to identify issues with slope, planarity, elevation or drainage. Additionally, unless the surfacing contractor is also responsible for the paving, it is inappropriate to hold that contractor responsible for such issues if the pavement has not been tested and corrected prior to surfacing. It is highly recommended that track projects include testing to demonstrate adherence to specifications and, if necessary, corrective work to achieve compliance at the cost of the asphalt contractor or the general contractor. These tests should be completed prior to the installation of track surfacing. An independent testing firm should check the dimensions, slopes and tolerances of the asphalt pavement. Results confirming compliance should be submitted to the surfacing contractor and owner prior to mobilization. If the initial assessment shows noncompliance, the necessary corrective work should be executed and the pavement retested prior to the commencement of surfacing. A few more points to consider:

- The asphalt mix design should be approved by the design professional, the surfacing contractor and the asphalt contractor. Tests that verify that the mix, as installed, meets the approved design should be provided.
- The surface course of asphalt shall be installed so that on a localized level, there shall be no bumps or depressions beneath a 10' straight edge exceeding 1/4 of an inch. Particular attention should be paid to seams and joints in the asphalt surface. The intent is to ensure the safety of the athlete and provide an even surface. Since the resilient running surface to be installed over the asphalt has tolerances with respect to thickness, it is important that the asphalt surface meets the tolerance described above. It is important to remember that the use of extra track surfacing materials to correct planarity deficiencies is not recommended.
- The installed pavement must be at the correct elevation at all curbs, channel drains and other adjoining surfaces. Incorrect areas should be repaired or reinstalled to correct elevations.
- All bond breakers, such as petroleum-based products like oil, hydraulic fluid and gasoline, must be removed.
- The asphalt should be flooded to check for low areas and surface drainage. Out-of-tolerance areas should be repaired or reinstalled.
- Prior to surfacing, the asphalt should be cured in accordance with the recommendations of the specification writer and the manufacturer of the surfacing system. This is typically 28 days based on depth of asphalt and oil content.
- If the track requires a certification that includes a topographic survey, the survey should be performed prior to surfacing.

Corrective work should be accomplished using appropriate methods, such as:

- For high spots:
 - grinding
 - heating and rolling (only if approved by the surfacing contractor)
- For low spots:
 - removing and replacing
 - keying into existing asphalt
 - hiring the surfacing contractor to apply leveling materials approved by the surfacing manufacturer

Because of the relatively rigid specifications and tolerances for asphalt pavements intended for track installations, it is recommended that only asphalt paving contractors with prior track building experience be considered.

4.2.4 Concrete Guidelines

Concrete is the product of mixing crushed stone or gravel and sand with Portland cement, making it an extremely rigid pavement. As such, it is also exceptionally durable, meaning low maintenance costs after initial construction, though initial cost may be somewhat higher than that for asphalt due to both installation and additional preparation involved with applying surface materials to concrete. If problems occur, concrete is more difficult to repair than asphalt.

Because of its durability, concrete may be useful on a site with poor soil or in a geographic region with extreme freeze-thaw cycles. Concrete pavement will be resistant to settling, heaving and vegetation encroachment; however, some cracking can be expected.

Depending on site conditions, various kinds of concrete pavement may be suitable for running tracks: nonreinforced concrete, reinforced concrete and post-tensioned concrete.

4.2.4a Materials

Concrete should comply with the standards of industry sources such as ASTM International or the American Concrete Institute. These standards address blending, mix percentages and similar issues. Depending on the size and the complexity of the facility under construction, the construction documents may detail the characteristics of the concrete to be used. The aggregate used in the concrete mix should be clean, hard, durable and graded within specified limits.

4.2.4b Reinforcement

Reinforcement materials are commonly incorporated in concrete construction to provide additional stabilization and crack resistance. Proper reinforcement will extend the life